

Sunline Filter

Component Design Document

1 Description

Sunline filter. Estimates the sun direction in the body frame, the body rate, and the coarse sun sensor intensity bias from the coarse sun sensor cosines and the body rate with a square root unscented Kalman filter, and publishes the estimate for the guidance algorithms. A diagnostics packet with the full filter state, covariance, and residuals is produced at a commanded period, and the pending measurements can be cleared by command. The filter keeps its own time base, which restarts at the tick after it is built or reset, and takes the time of each measurement from its data product. Wraps the SunlineFilterAlgorithm C++ algorithm via its C shim.

2 Requirements

No requirements have been specified for this component.

3 Design

3.1 At a Glance

Below is a list of useful parameters and statistics that give a quick look into the makeup of the component.

- **Execution** - *passive*
- **Number of Connectors** - 10
- **Number of Invokee Connectors** - 4
- **Number of Invoker Connectors** - 6
- **Number of Generic Connectors** - *None*
- **Number of Generic Types** - *None*
- **Number of Unconstrained Arrayed Connectors** - *None*
- **Number of Commands** - 2
- **Number of Parameters** - 13
- **Number of Events** - 3
- **Number of Faults** - *None*
- **Number of Data Products** - 1
- **Number of Data Dependencies** - 2
- **Number of Packets** - 1

3.2 Diagram

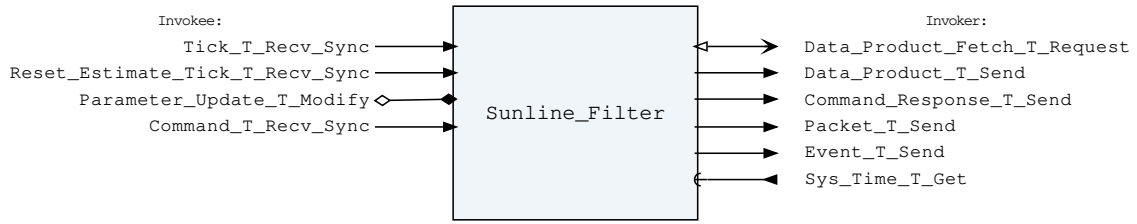


Figure 1: Sunline Filter component diagram.

3.3 Connectors

Below are tables listing the component's connectors.

3.3.1 Invokee Connectors

The following is a list of the component's *invokee* connectors:

Table 1: Sunline Filter Invokee Connectors

Name	Kind	Type	Return_Type	Count
Tick_T_Recv_Sync	recv_sync	Tick.T	-	1
Reset_Estimate_Tick_T_Recv_Sync	recv_sync	Tick.T	-	1
Parameter_Update_T_Modify	modify	Parameter_Update.T	-	1
Command_T_Recv_Sync	recv_sync	Command.T	-	1

Connector Descriptions:

- **Tick_T_Recv_Sync** - Run the filter up to the current time, folding in the coarse sun sensor and body rate readings that are new since the last tick.
- **Reset_Estimate_Tick_T_Recv_Sync** - Re-seed the filter state and covariance from the configured initial values and clear the pending measurements and residuals. The filter's time base restarts at the next tick.
- **Parameter_Update_T_Modify** - The parameter update connector.
- **Command_T_Recv_Sync** - This is the command receive connector.

3.3.2 Invoker Connectors

The following is a list of the component's *invoker* connectors:

Table 2: Sunline Filter Invoker Connectors

Name	Kind	Type	Return_Type	Count
Data_Product_Fetch_T_Request	request	Data_Product_Fetch.T	Data_Product_Return.T	1
Data_Product_T_Send	send	Data_Product.T	-	1
Command_Response_T_Send	send	Command_Response.T	-	1

Packet_T_Send	send	Packet.T	-	1
Event_T_Send	send	Event.T	-	1
Sys_Time_T_Get	get	-	Sys_Time.T	1

Connector Descriptions:

- **Data_Product_Fetch_T_Request** - Fetch a data product item from the database.
- **Data_Product_T_Send** - The data product invoker connector
- **Command_Response_T_Send** - This connector is used to register and respond to the component's commands.
- **Packet_T_Send** - The diagnostics packet is sent out of this connector.
- **Event_T_Send** - Events are sent out of this connector.
- **Sys_Time_T_Get** - The system time is retrieved via this connector.

3.4 Interrupts

This component contains no interrupts.

3.5 Initialization

Below are details on how the component should be initialized in an assembly.

3.5.1 Component Instantiation

This component contains no instantiation parameters in its discriminant.

3.5.2 Component Base Initialization

This component contains no base class initialization, meaning there is no `init_Base` subprogram for this component.

3.5.3 Component Set ID Bases

This component contains commands, events, packets, faults, or data products that require a base identifier to be set at initialization. The `set_Id_Bases` procedure must be called with the following parameters:

Table 3: Sunline Filter Set Id Bases Parameters

Name	Type
Command_Id_Base	Command_Types.Command_Id_Base
Data_Product_Id_Base	Data_Product_Types.Data_Product_Id_Base
Event_Id_Base	Event_Types.Event_Id_Base
Packet_Id_Base	Packet_Types.Packet_Id_Base
Parameter_Id_Base	Parameter_Types.Parameter_Id_Base

Parameter Descriptions:

- **Command_Id_Base** - The value at which the component's command identifiers begin.
- **Data_Product_Id_Base** - The value at which the component's data product identifiers begin.
- **Event_Id_Base** - The value at which the component's event identifiers begin.
- **Packet_Id_Base** - The value at which the component's unresolved packet identifiers begin.

- **Parameter_Id_Base** - The value at which the component's parameter identifiers begin.

3.5.4 Component Map Data Dependencies

This component contains data dependencies that need their IDs mapped to valid data product IDs at initialization. Each data dependency also needs a stale limit provided in microseconds. If the timestamp of the data dependency is older than the reference time (usually the current time) minus the stale limit then the data dependency is considered too old to safely use. The `map_Data_Dependencies` procedure must be called with the following parameters:

Table 4: Sunline Filter Map Data Dependencies Parameters

Name	Type
Body_Rate_Id	Data_Product_Types.Data_Product_Id
Body_Rate_Stale_Limit	Ada.Real_Time.Time_Span
Css_Cosines_Id	Data_Product_Types.Data_Product_Id
Css_Cosines_Stale_Limit	Ada.Real_Time.Time_Span

Parameter Descriptions:

- **Body_Rate_Id** - The external data product ID used to fetch the value for the 'Body_Rate' data dependency.
- **Body_Rate_Stale_Limit** - The stale limit used for the 'Body_Rate' data dependency. If the fetched data product is older than the given `Time_Span` then it is deemed stale and unfit for use by the component.
- **Css_Cosines_Id** - The external data product ID used to fetch the value for the 'Css_Cosines' data dependency.
- **Css_Cosines_Stale_Limit** - The stale limit used for the 'Css_Cosines' data dependency. If the fetched data product is older than the given `Time_Span` then it is deemed stale and unfit for use by the component.

3.5.5 Component Implementation Initialization

The calling of this implementation class initialization procedure is mandatory. Initializes the sunline filter with the default parameter values, which seed the filter state and covariance. The diagnostics packet is off until its period is commanded. The `init` subprogram takes no parameters.

3.6 Commands

Commands for the Sunline Filter component.

Table 5: Sunline Filter Commands

Local ID	Command Name	Argument Type
0	Set_Diagnostics_Packet_Period	Packed_U16.T
1	Reset_Measurements	-

Command Descriptions:

- **Set_Diagnostics_Packet_Period** - Set the period of the diagnostics packet in ticks. Zero turns the packet off.
- **Reset_Measurements** - Clear the pending measurements and residuals at the next tick, keeping the filter state and covariance. The filter's time base restarts at that tick. The filter drops a bad measurement on its own, so this is only needed for testing.

3.7 Parameters

Parameters for the Sunline Filter component. The process noise and the initial covariance are given by their diagonals, since the full matrices do not fit in a parameter.

Table 6: Sunline Filter Parameters

Local ID	Parameter Name	Type	Default Value
0x0000 (0)	Alpha	Packed_F64.T	(Value=>0.02)
0x0001 (1)	Beta	Packed_F64.T	(Value=>2.0)
0x0002 (2)	Process_Noise_Diagonal	Packed_F64x7. T	[1.0E-12, 1.0E-12, 1.0E-12, 1.0E-16, 1.0E-16, 1.0E-16,1. 0E-10]
0x0003 (3)	Initial_State	Packed_F64x7. T	[0.0,0.0,1.0, 0.0,0.0,0.0, 1.0]
0x0004 (4)	Initial_Covariance_Diagonal	Packed_F64x7. T	[0.1,0.1,0.1, 0.001,0.001, 0.001,0.5]
0x0005 (5)	Bias_Lower_Bound	Packed_F64.T	(Value=>0.5)
0x0006 (6)	Bias_Upper_Bound	Packed_F64.T	(Value=>1.5)
0x0007 (7)	Css_N_Hat_B	Packed_ F32x24.T	[+0.70710678, -0.50000000, +0.50000000, +0.70710678, -0.50000000, -0.50000000, +0.70710678, +0.50000000, -0.50000000, +0.70710678, +0.50000000, -0.50000000, +0.70710678, +0.50000000, -0.70710678, +0.00000000, +0.70710678, -0.70710678, +0.70710678, +0.00000000, -0.70710678, +0.00000000, -0.70710678, -0.70710678, -0.70710678, +0.00000000]
0x0008 (8)	Css_Scale_Factor	Packed_F64x8. T	[1.0,1.0,1.0, 1.0,1.0,1.0, 1.0,1.0]
0x0009 (9)	Number_Of_Css	Packed_U32.T	(Value=>8)
0x000a (10)	Sensor_Threshold	Packed_F64.T	(Value=>0.0)
0x000b (11)	Css_Measurement_Noise_Std	Packed_F64.T	(Value=>0.01)

0x000c (12)	Gyro_Measurement_Noise_Std	Packed_F64.T	(Value=>0.001)
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Parameter Descriptions:

- **Alpha** - [-] Sigma point spread. Must be in (0, 1].
- **Beta** - [-] Prior knowledge tunable. Must be in [0, 2].
- **Process_Noise_Diagonal** - [-] Diagonal of the N x N process noise covariance, one entry per state: the sun direction, the body rate, and the intensity bias. Every entry must be finite and not negative.
- **Initial_State** - [-] State the filter is seeded with: the sun direction in the body frame, the body rate in rad/s, and the intensity bias. Must be finite. The seed is taken when the filter is built and at the estimate reset; a parameter update alone keeps the running estimate.
- **Initial_Covariance_Diagonal** - [-] Diagonal of the N x N covariance the filter is seeded with, one entry per state. Every entry must be finite and not negative. The seed is taken when the filter is built and at the estimate reset; a parameter update alone keeps the running estimate.
- **Bias_Lower_Bound** - [-] Lower clamp on the intensity bias state. Must be greater than zero and less than the upper bound.
- **Bias_Upper_Bound** - [-] Upper clamp on the intensity bias state. Must be greater than the lower bound.
- **Css_N_Hat_B** - [-] Boresight unit vector of each coarse sun sensor slot in body frame components, three per sensor in row major order. The first configured number of sensors must be unit vectors to within 1e-3.
- **Css_Scale_Factor** - [-] Calibration scale factor of each coarse sun sensor slot. One means the nominal response. Every entry must not be negative.
- **Number_Of_Css** - [-] Number of coarse sun sensors in use, the first that many slots. Must be in [1, 8].
- **Sensor_Threshold** - [-] Cosine value above which a sensor counts as active in a measurement. Must not be negative.
- **Css_Measurement_Noise_Std** - [-] Standard deviation of the coarse sun sensor measurement noise. Must not be negative.
- **Gyro_Measurement_Noise_Std** - [rad/s] Standard deviation of the gyro measurement noise. Must not be negative.

3.8 Events

Events for the Sunline Filter component.

Table 7: Sunline Filter Events

Local ID	Event Name	Parameter Type
0	Diagnostics_Packet_Period_Set	Packed_U16.T
1	Measurements_Reset	-
2	Invalid_Command_Received	Invalid_Command_Info.T

Event Descriptions:

- **Diagnostics_Packet_Period_Set** - The diagnostics packet period was set via command, in ticks. Zero turns the packet off.

- **Measurements_Reset** - A command was received to clear the pending measurements and residuals at the next tick.
- **Invalid_Command_Received** - A command was received with invalid arguments.

3.9 Data Products

Data products for the Sunline Filter component.

Table 8: Sunline Filter Data Products

Local ID	Data Product Name	Type
0x0000 (0)	Sun_Direction_Estimate	Nav_Att_Output.T

Data Product Descriptions:

- **Sun_Direction_Estimate** - Estimated sun direction in the body frame and body rate, stamped with the time the filter advanced to. The attitude is not estimated by this filter and is zero.

3.10 Data Dependencies

Data dependencies for the Sunline Filter component. Each product is fed to the filter as a measurement when its timestamp has advanced since the last one consumed, so the timestamp is the measurement time.

Table 9: Sunline Filter Data Dependencies

Data Dependency Name	Type
Body_Rate	Packed_F32x3.T
Css_Cosines	Css_Sensor_Values.T

Data Dependency Descriptions:

- **Body_Rate** - [rad/s] Selected body rate in body frame components, the gyro measurement of the filter.
- **Css_Cosines** - [-] Corrected cosine value of each coarse sun sensor slot, the sun sensor measurement of the filter. Only the first configured number of sensors is read.

3.11 Packets

Packets for the Sunline Filter component.

Table 10: Sunline Filter Packets

Local ID	Packet Name	Type
0x0000 (0)	Filter_Diagnostics	Sunline_Filter_Output.T

Packet Descriptions:

- **Filter_Diagnostics** - The full filter snapshot after the update, the state, the covariance, and the residuals of each measurement kind. Sent every diagnostics packet period, in ticks, as set by command.

3.12 Faults

The Sunline Filter component has no faults.

4 Unit Tests

The following section describes the unit test suites written to test the component.

4.1 *Sunline_Filter_Tests* Test Suite

This is a unit test suite for the Sunline Filter component

Test Descriptions:

- **Test** - Feed the sun sensor cosines and body rate of a slow rotation and check the sun direction, rate, and bias estimates converge to the truth while the covariance shrinks, as in the algorithm's Python reference test, to ensure the Ada to C to C++ integration is sound.
- **Test_Propagation** - Ensure products whose timestamps have not advanced are not fed to the filter, so the sun direction is propagated with the seeded rate and the covariance grows under the process noise.
- **Test_Reading_Before_Time_Base** - Ensure products stamped before the time base restarted by a reset are dropped, and later ones are applied.
- **Test_Time_Anomalies** - Ensure products stamped after the tick wait until the tick reaches them, and a tick before the time base restarts it, without either shutting out the readings that follow.
- **Test_Fewer_Sensors** - Ensure only the configured number of sensors is read and only those above the threshold count as active.
- **Test_Reset** - Ensure the estimate reset re-seeds the state and covariance while the commanded measurements reset keeps them.
- **Test_Diagnostics_Packet** - Ensure the diagnostics packet is sent every commanded number of ticks, that a period of zero turns it off, and that the command is reported.
- **Test_Parameter_Update** - Ensure a parameter update reaches the filter while keeping the current estimate.
- **Test_Invalid_Parameter** - Ensure a staged configuration the algorithm would reject is refused at validation.
- **Test_Invalid_Command** - Ensure a malformed command is reported rather than acted on.
- **Test_Invalid_Data_Dependency** - Ensure a data dependency with the wrong identifier is treated as a wiring defect and fails the tick's assertion.

5 Appendix

5.1 Preamble

This component contains no preamble code.

5.2 Packed Types

The following section outlines any complex data types used in the component in alphabetical order. This includes packed records and packed arrays that might be used as connector types, command arguments, event parameters, etc..

Command.T:

Generic command packet for holding arbitrary commands

Table 11: Command Packed Record : 2080 bits (*maximum*)

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Command_Header.T	-	40	0	39	-
Arg_Buffer	Command_Types. Command_Arg_Buffer_Type	-	2040	40	2079	Header.Arg_Buffer_Length

Field Descriptions:

- **Header** - The command header
- **Arg_Buffer** - A buffer that contains the command arguments

Command_Header.T:

Generic command header for holding arbitrary commands

Table 12: Command_Header Packed Record : 40 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Source_Id	Command_Types. Command_Source_Id	0 to 65535	16	0	15
Id	Command_Types. Command_Id	0 to 65535	16	16	31
Arg_Buffer_Length	Command_Types. Command_Arg_Buffer_Length_Type	0 to 255	8	32	39

Field Descriptions:

- **Source_Id** - The source ID. An ID assigned to a command sending component.
- **Id** - The command identifier
- **Arg_Buffer_Length** - The number of bytes used in the command argument buffer

Command_Response.T:

Record for holding command response data.

Table 13: Command_Response Packed Record : 56 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Source_Id	Command_Types.Command_Source_Id	0 to 65535	16	0	15

Registration_Id	Command_Types.Command_Registration_Id	0 to 65535	16	16	31
Command_Id	Command_Types.Command_Id	0 to 65535	16	32	47
Status	Command_Enums.Command_Response_Status.E	0 => Success 1 => Failure 2 => Id_Error 3 => Validation_Error 4 => Length_Error 5 => Dropped 6 => Register 7 => Register_Source	8	48	55

Field Descriptions:

- **Source_Id** - The source ID. An ID assigned to a command sending component.
- **Registration_Id** - The registration ID. An ID assigned to each registered component at initialization.
- **Command_Id** - The command ID for the command response.
- **Status** - The command execution status.

Css_Sensor_Values.T:

Corrected CSS sensor cosine values for the 8 physical CSS ADC channels.

Table 14: Css_Sensor_Values Packed Record : 512 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Data	Packed_F64x8.T	-	512	0	511

Field Descriptions:

- **Data** - *No description provided.*

Data_Product.T:

Generic data product packet for holding arbitrary data types

Table 15: Data_Product Packed Record : 1096 bits (*maximum*)

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Data_Product_Header.T	-	72	0	71	-
Buffer	Data_Product_Types.Data_Product_Buffer_Type	-	1024	72	1095	Header.Buffer_Length

Field Descriptions:

- **Header** - The data product header

- **Buffer** - A buffer that contains the data product type

Data_Product_Fetch.T:

A packed record which holds information for a data product request.

Table 16: Data_Product_Fetch Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Data_Product_Types. Data_Product_Id	0 to 65535	16	0	15

Field Descriptions:

- **Id** - The data product identifier

Data_Product_Header.T:

Generic data_product packet for holding arbitrary data_product types

Table 17: Data_Product_Header Packed Record : 72 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	48	0	47
Id	Data_Product_Types. Data_Product_Id	0 to 65535	16	48	63
Buffer_Length	Data_Product_ Types.Data_Product_ Buffer_Length_Type	0 to 128	8	64	71

Field Descriptions:

- **Time** - The timestamp for the data product item.
- **Id** - The data product identifier
- **Buffer_Length** - The number of bytes used in the data product buffer

Data_Product_Return.T:

This record holds data returned from a data product fetch request.

Table 18: Data_Product_Return Packed Record : 1104 bits (*maximum*)

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
The_ Status	Data_ Product_ Enums. Fetch_ Status.E	0 => Success 1 => Not_Available 2 => Id_Out_Of_Range	8	0	7	-
The_ Data_ Product	Data_ Product. T	-	1096	8	1103	-

Field Descriptions:

- **The_Status** - A status relating whether or not the data product fetch was successful or not.
- **The_Data_Product** - The data product item returned.

Event.T:

Generic event packet for holding arbitrary events

Table 19: Event Packed Record : 328 bits (*maximum*)

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Event_Header.T	-	72	0	71	-
Param_Buffer	Event_Types. Parameter_ Buffer_Type	-	256	72	327	Header.Param_ Buffer_Length

Field Descriptions:

- **Header** - The event header
- **Param_Buffer** - A buffer that contains the event parameters

Event_Header.T:

Generic event packet for holding arbitrary events

Table 20: Event_Header Packed Record : 72 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	48	0	47
Id	Event_Types.Event_ Id	0 to 65535	16	48	63
Param_Buffer_Length	Event_Types. Parameter_Buffer_ Length_Type	0 to 32	8	64	71

Field Descriptions:

- **Time** - The timestamp for the event.
- **Id** - The event identifier
- **Param_Buffer_Length** - The number of bytes used in the param buffer

Invalid_Command_Info.T:

Record for holding information about an invalid command

Table 21: Invalid_Command_Info Packed Record : 112 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Command_Types. Command_Id	0 to 65535	16	0	15
Errant_Field_ Number	Interfaces. Unsigned_32	0 to 4294967295	32	16	47
Errant_Field	Basic_Types.Poly_ Type	-	64	48	111

Field Descriptions:

- **Id** - The command Id received.
- **Errant_Field_Number** - The field that was invalid. 1 is the first field, 0 means unknown field, 2^{32} means that the length field of the command was invalid.
- **Errant_Field** - A polymorphic type containing the bad field data, or length when Errant_Field_Number is 2^{32} .

Nav_Att_Output.T:

Table 22: Nav_Att_Output Packed Record : 352 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time_Tag	Long_Float	$-\infty$ to ∞	64	0	63
Sigma_Bn	Packed_F32x3.T	-	96	64	159
Omega_Bn_B	Packed_F32x3.T	-	96	160	255
Veh_Sun_Pnt_Bdy	Packed_F32x3.T	-	96	256	351

Field Descriptions:

- **Time_Tag** - [s] Time tag of navigation solution
- **Sigma_Bn** - MRP attitude of body w.r.t. inertial frame
- **Omega_Bn_B** - [rad/s] Angular velocity of body w.r.t. inertial, body frame components
- **Veh_Sun_Pnt_Bdy** - Sun direction unit vector in body frame

Packed_F32x24.T:

Packed 24 element vector of 32-bit floats. Holds a table of 8 three-component vectors in row major order.

Table 23: Packed_F32x24 Packed Array : 768 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Short_Float	$-3.40282e+38$ to $3.40282e+38$	32	24	768

Packed_F32x3.T:

Packed 3 element vector of 32-bit floats.

Table 24: Packed_F32x3 Packed Array : 96 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Short_Float	$-3.40282e+38$ to $3.40282e+38$	32	3	96

Packed_F64.T:

Single component record for holding packed 64-bit floating point number.

Table 25: Packed_F64 Packed Record : 64 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Long_Float	-inf to inf	64	0	63

Field Descriptions:

- **Value** - The 64-bit floating point number.

Packed_F64x3.T:

Packed 3 element vector of 64-bit floats.

Table 26: Packed_F64x3 Packed Array : 192 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Long_Float	-inf to inf	64	3	192

Packed_F64x49.T:

Packed 49 element vector of 64-bit floats.

Table 27: Packed_F64x49 Packed Array : 3136 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Long_Float	-inf to inf	64	49	3136

Packed_F64x7.T:

Packed 7 element vector of 64-bit floats.

Table 28: Packed_F64x7 Packed Array : 448 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Long_Float	-inf to inf	64	7	448

Packed_F64x8.T:

Packed 8 element vector of 64-bit floats.

Table 29: Packed_F64x8 Packed Array : 512 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Long_Float	-inf to inf	64	8	512

Packed_U16.T:

Single component record for holding packed unsigned 16-bit value.

Table 30: Packed_U16 Packed Record : 16 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Interfaces. Unsigned_16	0 to 65535	16	0	15

Field Descriptions:

- **Value** - The 16-bit unsigned integer.

Packed_U32.T:

Single component record for holding packed unsigned 32-bit value.

Table 31: Packed_U32 Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Interfaces. Unsigned_32	0 to 4294967295	32	0	31

Field Descriptions:

- **Value** - The 32-bit unsigned integer.

Packet.T:

Generic packet for holding arbitrary data

Table 32: Packet Packed Record : 8128 bits (*maximum*)

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Packet_ Header.T	-	96	0	95	-
Buffer	Packet_ Types.Packet_ Buffer_Type	-	8032	96	8127	Header. Buffer_Length

Field Descriptions:

- **Header** - The packet header
- **Buffer** - A buffer that contains the packet data

Packet_Header.T:

Generic packet header for holding arbitrary data

Table 33: Packet_Header Packed Record : 96 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	48	0	47
Id	Packet_Types. Packet_Id	0 to 65535	16	48	63

Sequence_Count	Packet_Types. Sequence_Count_Mod_ Type	0 to 16383	16	64	79
Buffer_Length	Packet_Types. Packet_Buffer_ Length_Type	0 to 1004	16	80	95

Field Descriptions:

- **Time** - The timestamp for the packet item.
- **Id** - The packet identifier
- **Sequence_Count** - Packet Sequence Count
- **Buffer_Length** - The number of bytes used in the packet buffer

Parameter.T:

Generic parameter packet for holding a generic parameter

Table 34: Parameter Packed Record : 792 bits (*maximum*)

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Parameter_ Header.T	-	24	0	23	-
Buffer	Parameter_ Types. Parameter_ Buffer_Type	-	768	24	791	Header.Buffer_ Length

Field Descriptions:

- **Header** - The parameter header
- **Buffer** - A buffer that contains the parameter type

Parameter_Header.T:

Generic parameter header for holding arbitrary parameters

Table 35: Parameter_Header Packed Record : 24 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Parameter_Types. Parameter_Id	0 to 65535	16	0	15
Buffer_Length	Parameter_Types. Parameter_Buffer_ Length_Type	0 to 96	8	16	23

Field Descriptions:

- **Id** - The parameter identifier
- **Buffer_Length** - The number of bytes used in the parameter type buffer

Parameter_Update.T:

A record intended to be used as a provide/modify connector type for updating/fetching parameters.

Table 36: Parameter_Update Packed Record : 824 bits (*maximum*)

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Table_Id	Parameter_Types. Parameter_Table_Id	0 to 65535	16	0	15	–
Operation	Parameter_Enums. Parameter_Operation_Type.E	0 => Stage 1 => Update 2 => Fetch 3 => Validate	8	16	23	–
Status	Parameter_Enums. Parameter_Update_Status.E	0 => Success 1 => Id_Error 2 => Validation_Error 3 => Length_Error	8	24	31	–
Param	Parameter.T	-	792	32	823	–

Field Descriptions:

- **Table_Id** - The ID for the table that contains this parameter
- **Operation** - The parameter operation to perform.
- **Status** - The parameter return status.
- **Param** - The parameter that has been updated or fetched.

Sunline_Filter_Css_Residuals.T:

Residuals of the coarse sun sensor measurement from one sunline filter update. Maps to SunlineCss-ResidualsOutput_c in the sunlineFilter C shim.

Table 37: Sunline_Filter_Css_Residuals Packed Record : 1576 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Valid	Boolean	0 => False 1 => True	8	0	7
Number_Of_Active_Css	Interfaces. Integer_32	-2147483648 to 2147483647	32	8	39
Observation	Packed_F64x8.T	-	512	40	551
Pre_Fit	Packed_F64x8.T	-	512	552	1063
Post_Fit	Packed_F64x8.T	-	512	1064	1575

Field Descriptions:

- **Valid** - Whether a CSS measurement fired this update. The other fields are meaningful only when it did. Mirrors the C99 bool in the shim's struct.
- **Number_Of_Active_Css** - [-] Number of sensors above the sensor threshold. Only that many entries of each array are meaningful.
- **Observation** - [-] Measured cosine values.
- **Pre_Fit** - [-] Residuals before the update.
- **Post_Fit** - [-] Residuals after the update.

Sunline_Filter_Output.T:

Snapshot of the sunline filter after an update. Maps to SunlineFilterOutput_c in the sunlineFilter C shim, which returns it by value.

Table 38: Sunline_Filter_Output Packed Record : 5744 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Filter_State	Sunline_Filter_State_Output.T	-	3584	0	3583
Css_Residuals	Sunline_Filter_Css_Residuals.T	-	1576	3584	5159
Rate_Residuals	Sunline_Filter_Rate_Residuals.T	-	584	5160	5743

Field Descriptions:

- **Filter_State** - The state and covariance after the update.
- **Css_Residuals** - Coarse sun sensor residuals of this update.
- **Rate_Residuals** - Gyro rate residuals of this update.

Sunline_Filter_Rate_Residuals.T:

Residuals of the gyro rate measurement from one sunline filter update. Maps to SunlineRateResidualsOutput_c in the sunlineFilter C shim.

Table 39: Sunline_Filter_Rate_Residuals Packed Record : 584 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Valid	Boolean	0 => False 1 => True	8	0	7
Observation	Packed_F64x3.T	-	192	8	199
Pre_Fit	Packed_F64x3.T	-	192	200	391
Post_Fit	Packed_F64x3.T	-	192	392	583

Field Descriptions:

- **Valid** - Whether a rate measurement fired this update. The other fields are meaningful only when it did. Mirrors the C99 bool in the shim's struct.
- **Observation** - [rad/s] Measured body rate.
- **Pre_Fit** - [rad/s] Residuals before the update.
- **Post_Fit** - [rad/s] Residuals after the update.

Sunline_Filter_State_Output.T:

State and covariance of the sunline filter after an update. Maps to SunlineFilterStateOutput_c in the sunlineFilter C shim.

Table 40: Sunline_Filter_State_Output Packed Record : 3584 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
State	Packed_F64x7.T	-	448	0	447
Covariance	Packed_F64x49.T	-	3136	448	3583

Field Descriptions:

- **State** - [-] Estimated state: the sun direction in the body frame, the body rate in rad/s, and the CSS intensity bias.
- **Covariance** - [-] N x N state covariance in row major order.

Sys_Time.T:

A record which holds a time stamp using EMA VTC format including seconds and subseconds since epoch (1-5-1980 to 1-6-1980 midnight).

Table 41: Sys_Time Packed Record : 48 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Seconds	Interfaces. Unsigned_32	0 to 4294967295	32	0	31
Subseconds	Interfaces. Unsigned_16	0 to 65535	16	32	47

Field Descriptions:

- **Seconds** - The number of seconds elapsed since epoch.
- **Subseconds** - The number of $1/(2^{16})$ sub-seconds.

Tick.T:

The tick datatype used for periodic scheduling. Included in this type is the Time associated with a tick and a count.

Table 42: Tick Packed Record : 80 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	48	0	47
Count	Interfaces. Unsigned_32	0 to 4294967295	32	48	79

Field Descriptions:

- **Time** - The timestamp associated with the tick.
- **Count** - The cycle number of the tick.

5.3 Enumerations

The following section outlines any enumerations used in the component.

Command_Enums.Command_Response_Status.E:

This status enumeration provides information on the success/failure of a command through the command response connector.

Table 43: Command_Response_Status Literals:

Name	Value	Description
Success	0	Command was passed to the handler and successfully executed.
Failure	1	Command was passed to the handler not successfully executed.
Id_Error	2	Command id was not valid.
Validation_Error	3	Command parameters were not successfully validated.
Length_Error	4	Command length was not correct.
Dropped	5	Command overflowed a component queue and was dropped.
Register	6	This status is used to register a command with the command routing system.
Register_Source	7	This status is used to register command sender's source id with the command router for command response forwarding.

Data_Product_Enums.Fetch_Status.E:

This status denotes whether a data product fetch was successful.

Table 44: Fetch_Status Literals:

Name	Value	Description
Success	0	The data product was returned successfully.
Not_Available	1	No data product is yet available for the provided id.
Id_Out_Of_Range	2	The data product id was out of range.

Parameter_Enums.Parameter_Operation_Type.E:

This enumeration lists the different parameter operations that can be performed.

Table 45: Parameter_Operation_Type Literals:

Name	Value	Description
Stage	0	Stage the parameter.
Update	1	All parameters are staged, it is ok to update all parameters now.
Fetch	2	Fetch the parameter.
Validate	3	Validate all the parameters.

Parameter_Enums.Parameter_Update_Status.E:

This status enumeration provides information on the success/failure of a parameter operation.

Table 46: Parameter_Update_Status Literals:

Name	Value	Description
Success	0	Parameter was successfully staged.
Id_Error	1	Parameter id was not valid.
Validation_Error	2	Parameter values were not successfully validated.
Length_Error	3	Parameter length was not correct.